

Auditing LLMs for Academic Citation Hallucinations in African American English

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Abstract

Large language models (LLMs) are increasingly used by students as informal research assistants, and prior work has shown that LLMs covertly discriminate against speakers of African American English (AAE). Separately, LLMs are known to hallucinate, especially when asked about academic references. We ask whether dialect of the query affects the rate or character of hallucinations in citation-grounded academic queries. We construct a paired dataset of 1200 prompts (600 pairs) in a 2×2 within-item design (dialect: SAE/AAE × source-citation type: fabricated/underspecified), pass each through GPT-5, and annotate every response for hallucinations with an LLM-as-a-judge pipeline. Pooled across conditions we observe a small AAE-directional effect on hallucination rate (+2.9 percentage points), which is not statistically significant under a paired McNemar test ($p \approx 0.20$). Citation type, however, dominates dialect: underspecified prompts hallucinate at ~81–82% versus ~47–52% on fabricated prompts ($\chi^2, p < 10^{-14}$ in both dialects). We complement the quantitative analysis with a reflexive thematic analysis of judge-evidence text, surfacing five themes that describe how the model fails: *hedge-then-fabricate*, *tacit ratification of ambiguity*, *real-authority anchoring*, *fabricated specificity*, and *confidence-outcome decoupling*. We release the annotated dataset and judge prompts, validate and release the dataset creation pipeline to enable future extensions, and conduct a power analysis indicating that detecting an effect of this size at $\alpha=0.05$ with 80% power would require approximately 1400 paired prompts.

Keywords

large language models, hallucination, African American English, dialect, fairness, audit

1 Introduction

Large language models (LLMs) such as GPT-5 are now embedded in everyday academic workflows: students paste citations into a chatbot to ask what a paper argues, what its methodology is, or whether it is relevant to a given topic. At the same time, two failure modes of LLMs have been independently documented. First, LLMs exhibit *covert* racial bias along dialectal lines: when prompted with text in African American English (AAE), models systematically generate more negative, more stereotypical, and lower-status outputs about the (implied) speaker than for matched Standard American English (SAE) prompts, and this effect persists in models that are explicitly aligned to be unbiased [1, 8, 10, 11]. Second, LLMs hallucinate [2, 12, 13] – they generate plausible-sounding but factually wrong or unfaithful content, which we anecdotally note and empirically find is particularly common in citation-related queries where the model is asked to summarize a paper it has never seen.

These two failure modes have, to our knowledge, not been studied jointly. LLMs are widely framed as democratizing access to education, and free or low-cost LLM tools are disproportionately

used by students who lack access to other forms of academic support. AAE is also disproportionately spoken by Black Americans, and historically by lower-income Black Americans [15]. If models are quietly less reliable – or hallucinate *differently* – when queried in AAE, then the very groups for whom the “democratization” framing is most salient may be receiving systematically worse information from the systems they rely on most heavily, in a way that is invisible to both the user and the model provider.

We run a paired audit. We construct 300 academic citation queries with *fabricated* (non-existent) references and 300 with *underspecified* (real but obscure) references. Each is rendered in both SAE and AAE using a rule-based dialect translator with manual quality control, yielding 1200 prompts in a 2×2 within-item design. We pass each prompt through GPT-5 and annotate every response with an LLM-as-a-judge pipeline that follows the hallucination taxonomy of Huang et al. (2025) [12]. We then test for dialect effects with a paired McNemar test on the binary hallucination outcome, quantify case-level agreement with Cohen’s κ , and qualitatively analyze the judge-evidence column with a reflexive thematic analysis (Braun & Clarke [4]) to characterize *how* the model fails when it does.

Our contributions are: (1) an annotated dataset of 1200 paired SAE/AAE academic-citation prompts + responses from GPT-5, with hallucination type, subtype, and confidence annotations following Huang et al. (2025) [12]; (2) a validated pipeline for extending this dataset to additional models and domains; (3) a pilot analysis showing a small AAE-directional effect on hallucination rate in GPT-5 (**+2.9 percentage points pooled**, $p \approx 0.20$ paired McNemar), which is directional but underpowered; and (4) a reflexive thematic analysis identifying five qualitative themes that describe the model’s failure mode at the response level. We also discuss why the pipeline naturally extends to higher-stakes domains such as police transcript summarization, where directional effects of this kind would carry sharper civic consequences.

2 Related Work

Our project calls upon prior work in dialect bias in Natural Language Processing, the identification and generation of AAE text, and recent approaches to hallucination in LLMs.

2.1 Covert dialect-based discrimination in LLMs

The most direct precedent for this paper is Hofmann et al. [10], who use matched-guise probing to show that LLMs make markedly more negative implicit judgments about AAE-associated speakers than SAE-associated speakers, even when explicit prompts about “Black speakers” yield positive associations. The authors document downstream harms (employment, criminal sentencing) when LLMs are used as decision aids. Subsequent work has extended this finding to other dialect-conditioned harms [5, 6, 11]. Our study is closest in spirit to this line, but asks a question they do not: whether the factual and faithful reliability of LLM outputs (not just their

valence or stylistic register) varies by dialect. More recent work has established a framework for auditing LLMs for dialect-based harms [9].

2.2 Obtaining AAE data

Early work in AAE NLP relied on pre-existing corpora of naturally-occurring text, such as Twitter data [3], to study dialect-associated disparities in tasks like hate speech detection [16] and language identification. Prior work has also developed methods for identifying and classifying dialect features in model inputs and outputs, using both human-annotated corpora and automatic tools [5]. The VALUE benchmark and its multilingual extension Multi-VALUE [19, 20] translate between dialects using rule-based transformations grounded in linguistic features (e.g., habitual *be*, copula deletion, negation concord). AAVENUE [7] uses an LLM to translate SAE to AAE with few-shot prompting. While we acknowledge that LLM-based translations produce more fluent text, they have also been documented to introduce stereotypical features (profanity, informality) not actually associated with AAE. We use Multi-VALUE precisely to avoid propagating these stereotypes, and discuss the trade-off explicitly in §3.

2.3 Hallucination taxonomy and LLM-as-judge evaluation

We adopt a hallucination taxonomy from Huang et al. (2025) [12, 14]: specifically, a two-axis framework distinguishing factuality hallucinations (the output contradicts or fabricates external facts) from faithfulness hallucinations (the output contradicts the prompt or itself), with named subtypes. LLM-as-a-judge evaluation [17] has been shown to elicit over 80% agreement with expert human annotators; we anecdotally confirm performance for our specific task on a small sample of our data.

2.4 Research Gap

While prior work establishes the severity of dialect-based racial discrimination in LLMs, and separate work analyzes hallucination typology and prevalence in LLMs, to our knowledge these areas have not been combined. That is, we are unaware of any study that has analyzed whether racial dialect influences the character or frequency of factual and faithful inconsistencies in LLMs. We pilot an approach to answering this significant gap with a small but relevant setting: academic citations.

3 Methods

3.1 Design

We adopt a 2×2 within-item paired design. Each “item” is a single citation query (e.g., “I came across this citation: *Author (Year). Title. Journal, Vol(Issue), pages.* Can you summarize the paper’s main argument?”). Citations were generated by an LLM (GPT-5.4, online interface) using few-shot prompting with examples written by us (prompts included in supplementary material). Each item is rendered in two dialect conditions (SAE, AAE) and falls into one of two source-content conditions:

- **Fabricated** ($n=300$): the cited paper does not exist. LLM prompt generator used web search to verify citations were

absent from Google Scholar, Semantic Scholar, and Cross-Ref.

- **Underspecified** ($n=300$): the citation is to a real but obscure paper – typically published within the last ~2 years and/or with few citations – in the authors’ research areas (broadly, computer science / AI). Pilot work ($n=40$) showed that very popular real papers produced zero hallucinations in either dialect, which is uninformative.

This yields $300 \times 2 \times 2 = 1200$ total prompts.

3.2 SAE prompt generation

SAE prompts were generated with few-shot LLM prompting, using diverse contexts (e.g., “I found this in a syllabus for my X class”, “This reference was on an annotated bibliography for Y”) and varied request types (“Can you summarize the main argument?”, “What was the methodology?”). Bibliographic information (author names, year, title, venue, pagination) was authored by hand for the fabricated condition and copied verbatim from real venues for the underspecified condition. Full prompt generation prompts and the few-shot exemplars are available in our supplementary materials. The underspecified condition divided prompts equally among the following categories: {Author name degradation, Title degradation, Venue/metadata degradation, Conceptual misattribution, Mixed}. More information on these categories is available in Appendix A.

3.3 AAE translation: Multi-VALUE

Following Harvey et al. (2025) [9], we use Multi-VALUE [20] for SAE→AAE conversion of the surrounding question (e.g., “I came across this source while doing background research” → “I come across this here source while doing backgrounds researchs”).¹ The bibliographic block itself (author names, title, journal, year, pagination) is held fixed across dialects so that any difference in model output is attributable to the framing, not to the citation having been mangled. We note that although much of the past literature on dialect-based LLM audits uses natural AAE text written by native speakers [5, 6, 10, 11] or few-shot prompting of LLMs to translate between SAE and AAE [7, 18], we chose the rule-based translation system Multi-VALUE as we do not have the resources to obtain native speaker data and past LLM-based translations have been documented to introduce stereotyped features (profanity, exaggerated informality) not actually associated with AAE [5, 7, 9], which would confound any dialect effect with a separate “register effect.”

Manual QA. A co-author briefly hand-checked every translation for obvious errors such as placeholder translation text or corrupted citation strings. 16 of 600 were noted to contain such syntactic errors; these were minimally patched (typically a citation fix). We discuss the ecological-validity limitation of rule-based dialect generation in §6.

3.4 Model and inference

All 1200 prompts were submitted to GPT-5 via the online user interface. Each prompt was issued in an independent, anonymous session.

¹Specifically, we use the Python implementation value-nlp: <https://github.com/SALT-NLP/multi-value>

Table 1: Annotation codes mapped to the Huang et al. (2025) [12] taxonomy. [†] F3a extends the original taxonomy and has no direct counterpart in Huang et al.

Code	Huang et al. Type	Description
<i>Factuality Hallucination</i>		
F1	Entity-error	Wrong entity accepted without correction: author name, title, method name, or affiliation.
F2	Relation-error	Wrong relation accepted without correction: publication year, venue, or journal.
F3	Unverifiability	Specific paper content (findings, methodology, argument) asserted without adequate hedging; paper cannot be verified.
F3a [†]	Unverifiability	<i>Subtype of F3.</i> A real scholar’s identity attributed to a fictional author name.
F4	Overclaim	Paper’s scope, significance, or findings overstated beyond what the record supports.
<i>Faithfulness Hallucination</i>		
G1	Instruction Inconsistency	Model performed a different task than requested without acknowledgment.
G2	Context Inconsistency	Model contradicts or silently accepts erroneous information from the student prompt.
G3	Logical Inconsistency	Response contains internal contradictions.

3.5 Hallucination annotation pipeline

We use an LLM-as-a-judge pipeline keyed to the Huang et al. (2025) [12] taxonomy. Each response is labeled along four columns:

- **Hallucination** (Yes / No / No output);
- **Type**: comma-separated codes from {F1, F2, F3, F3a, F4, G1, G2, G3}, where Fx are factuality codes and Gx are faithfulness codes (Table 1);
- **Subtype label**: a free-text severity tag (e.g., “F3 – Severity 2: Moderately specific fabricated methodology”);
- **Confidence** of the judge: High / Medium / Low;
- **Evidence**: a 1–3 sentence rationale citing concrete spans of the response.

We use Claude Sonnet 4.6 as the judge for the fabricated condition (where the decision rule is essentially: “does the response confidently describe a paper that does not exist, or does it adequately hedge?”) and Claude Opus 4.7—a more powerful, but expensive, model—for the underspecified condition (where the judge must verify content against an actual paper, requiring more careful reading). The full judge system prompts are included in our supplementary material.

Validation. We hand-audited 20 randomly sampled judge outputs from each of the four cells (80 total). We found no clear hallucinations or inconsistencies in the judges’ own annotations; the judge confidence labels also tracked our hand judgments closely (high-confidence judgments were never wrong; medium-confidence judgments were occasionally borderline, but reflected legitimately

difficult decisions). We acknowledge that our validation methodology was not formal; §6 flags this as a limitation.

3.6 Statistical analysis

The primary test is a paired McNemar test on the binary hallucination outcome, separately within fabricated and underspecified conditions, and pooled. Because the paired design controls for item-level difficulty (the same fabricated citation is shown in both dialects), within-item discordances (*b* and *c* in the standard 2×2 paired table) carry the inferential weight. Marginal χ^2 is reported as a secondary check. We additionally compute Cohen’s κ to quantify dialect-level agreement on the same prompt; report Pearson χ^2 with Fisher’s exact OR for cross-category (fabricated vs. underspecified) comparisons within each dialect; tabulate Huang-code composition among hallucinated rows; and run a confidence-by-outcome χ^2 to assess whether the judge’s confidence ratings track factual accuracy. “No output” and ERROR rows² are excluded from paired tests, leaving $n_{\text{paired}}=286$ in the fabricated condition and 287 in the underspecified condition. All tests are two-sided; $\alpha=0.05$.

3.7 Reflexive thematic analysis

To complement the closed-set Huang taxonomy with an open description of *how* the model fails, we conducted a reflexive thematic analysis (RTA) of the judge-evidence column, following the six phases of Braun and Clarke [4]. (i) *Familiarisation*: we skimmed a sample of evidence-column entries and subtype-label, attending especially to recurring patterns of hedging, attribution, and authority. (ii) *Initial coding*: we generated descriptive codes from the evidence text (e.g., “opens with disclaimer then attributes content,” “never flags ambiguous citation,” “borrowed real-scholar identity,” “invented taxonomy with named categories,” “high confidence despite contradiction”). (iii) *Searching for themes*: we clustered codes by their central organizing concept rather than by topic, asking what story the model’s failure mode tells about how it manages epistemic risk. (iv) *Reviewing themes*: we tested each candidate theme against discordant cases (e.g., correctly-refused fabricated prompts) to ensure it earned its claim across the data. (v) *Defining and naming*: themes were named to surface the underlying mechanism (e.g., “hedge-then-fabricate” rather than “hedging issues”). (vi) *Producing the report*: each theme is presented with a definition, prevalence quantification, and one or two illustrative quotes drawn from the evidence column. To support the qualitative reading with prevalence numbers, we performed regex pattern counts on the evidence column and on the F-code distribution; these counts are descriptive, not constitutive of the themes themselves.

4 Results

4.1 Sample after exclusions

Of 1,200 total responses, 28 were excluded: 7 SAE-fabricated, 8 AAE-fabricated, 4 SAE-underspecified, and 9 AAE-underspecified. Exclusions were almost exclusively cases where we mis-clicked during response collection. Because we use a paired design, an exclusion in either dialect drops the entire item; this leaves $n=286$

²Generally, corrupted data due to author error

Table 2: Paired hallucination rates by condition. *b* = AAE-directional discordant items (SAE no, AAE yes); *c* = SAE-directional discordant items.

Condition	<i>n</i>	SAE%	AAE%	Δ pp	<i>b/c</i>	<i>p</i>
Fabricated	286	46.5	51.0	+4.5	58/45	0.24
Underspecified	287	80.8	82.2	+1.4	30/26	0.69
Pooled	573	63.7	66.7	+2.9	88/71	0.20

paired items in the fabricated condition and $n=287$ in the underspecified condition.

4.2 Headline numbers

Table 2 summarizes the paired hallucination rates and McNemar *p*-values. Across all conditions, AAE prompts produced numerically more hallucinations than SAE prompts, but the effect was not statistically significant in any cell.

4.3 Citation type dominates dialect

The largest effect in the dataset is not dialect but citation type. Within each dialect, fabricated and underspecified prompts produce dramatically different hallucination rates: SAE fabricated 47.1% vs. underspecified 81.1% ($\chi^2 = 72.5$, $p < 10^{-16}$, OR = 0.21); AAE fabricated 51.7% vs. underspecified 82.5% ($\chi^2 = 61.1$, $p < 10^{-14}$, OR = 0.23). Underspecified prompts trigger hallucinations roughly 1.7× more often than fully fabricated ones in both dialects. This effect dwarfs any dialect signal and inverts the intuitive ordering that fabricated prompts ought to be “obviously” harder to handle.

4.4 Paired agreement: equal marginals, divergent cases

Equal marginal rates do not imply equal case-level outcomes. On the same matched prompt, Cohen’s κ for SAE–AAE agreement on the binary hallucination outcome is $\kappa = 0.281$ (fabricated) and $\kappa = 0.352$ (underspecified) – both in the “fair to moderate” range. In the fabricated condition, both dialects hallucinate on 88 prompts, neither on 95, SAE-only on 45, and AAE-only on 58: the model disagrees with itself by dialect on 36% of prompts (103/286). In the underspecified condition the disagreement is smaller but still substantial: both 206, neither 25, SAE-only 26, AAE-only 30 (56/287, 19.5%). Translated procedurally, a user asking the same factual question in SAE versus AAE has roughly a one-in-five (underspecified) to one-in-three (fabricated) chance of receiving qualitatively different epistemic treatment, even when aggregate accuracy is matched.

4.5 Hallucination type composition

Figure 1 shows the distribution of hallucination types within hallucinated responses, presented as a grouped bar chart (codes are not mutually exclusive, so a single response may contribute to multiple bars). In the fabricated condition, the dominant code is **F3** (factual fabrication / unverifiable content); F3 alone or in combination accounts for nearly all hallucinations (99–100% of hallucinated responses in both dialects). The rare F3a code – assigning

the fabricated paper to a *real* scholar with the same surname (e.g., misattributing a fictional “Daniels & Ito” paper to media scholar Mizuko Ito) – appears 5 times in SAE-fabricated and 5 times in AAE-fabricated, i.e., perfectly symmetric. This is the most severe error class in our taxonomy and shows no dialect effect.

In the underspecified condition the picture is qualitatively different: F1 (entity error – typically wrong author or wrong first name) appears in 65.4% of SAE-underspecified hallucinations and 62.1% of AAE-underspecified hallucinations; F3 (fabricated content about a real paper) in 52.9% (SAE) and 61.3% (AAE); F2 (overclaim / scope error) in 42.5% in both dialects; with smaller contributions from F4 and G2. The most concrete dialect-level signal in the dataset is the +8.3 pp shift in F3 prevalence among AAE-underspecified hallucinations: when AAE prompts trigger underspecified hallucinations they skew further toward outright content fabrication relative to mere entity error. The paired contingency tables are shown in Figure 2.

4.6 Confidence calibration

Approximately 87% of judge labels are High-confidence; the remaining ~13% are Medium, with a single Low. A confidence-by-outcome χ^2 is statistically significant in all four cells, but the effect size is small: judge-rated High confidence drops only modestly when the response is hallucinated. In the fabricated condition, High confidence falls from 100% (correct responses) to 79% (SAE hallucinations) and 75% (AAE hallucinations). In the underspecified condition the relationship inverts: judges report *higher* High-confidence rates on hallucinated underspecified responses (87% SAE, 88% AAE) than on correct ones (75% SAE, 65% AAE), because hallucinated responses commit to a definite (if wrong) referent while correct responses often hedge the partial citation. Restricting all paired tests to High-confidence labels does not materially change any of the rates in Table 2.

4.7 Underspecified hallucination is high in both dialects

A finding worth flagging independently of the dialect comparison: in the underspecified condition GPT-5 hallucinates on roughly 80% of queries regardless of dialect. That is, when asked about a real but obscure paper, the modal behavior is to confidently confirm or fabricate incorrect attributions, methodology, findings, and arguments. The dialect effect we are searching for is small *relative to this floor*.

4.8 Reflexive thematic analysis

The RTA produced five themes. Each is presented with a definition, a prevalence anchor drawn from the evidence-column patterns, and one illustrative quote.

Theme 1: Hedge-then-Fabricate. The central organizing concept is that disclaimers function as decoration rather than as scope-control. The model opens with epistemic caution (e.g., “*I couldn’t find a publicly accessible full text of that exact article*”) but the hedge does not cover the rest of the response. Within paragraphs the framing slides from reconstruction to direct attribution: phrases like “*the article’s main argument is...*” or “*the authors suggest...*” appear

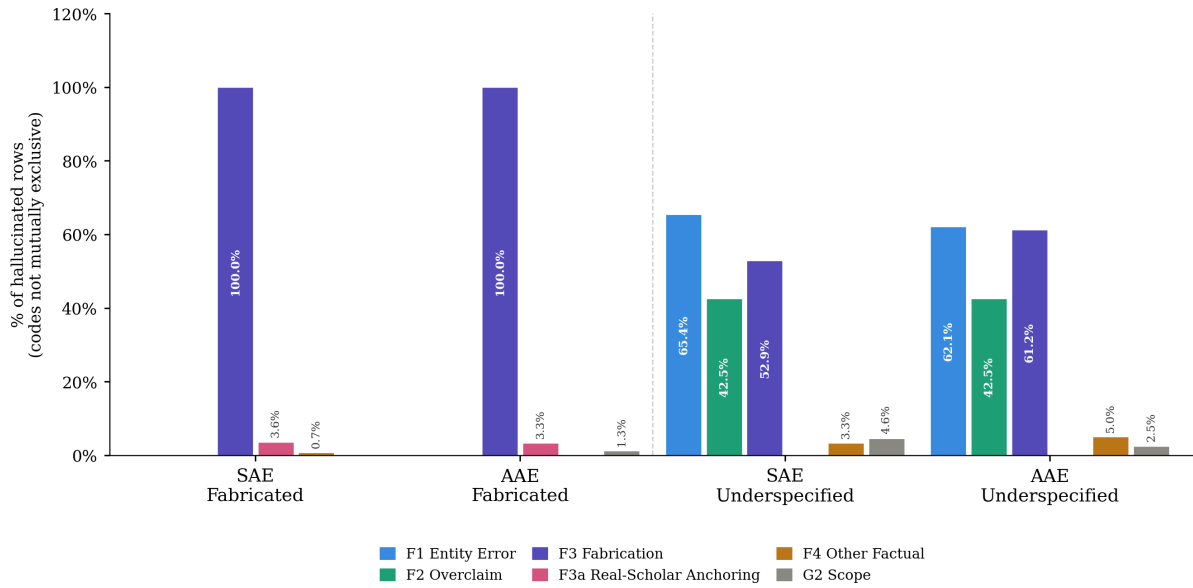


Figure 1: Hallucination type composition by condition.

Fabricated (n = 286)

	AAE: Yes	AAE: No	Total
SAE: Yes	88	45 ^(c)	133
SAE: No	58 ^(b)	95	153
Total	146	140	286

 $\kappa = 0.281$ · McNemar $p = 0.237$

Discordant: 103/286 (36%)

b = SAE no, AAE yes (AAE-directional)

Underspecified (n = 287)

	AAE: Yes	AAE: No	Total
SAE: Yes	206	26 ^(c)	232
SAE: No	30 ^(b)	25	55
Total	236	51	287

 $\kappa = 0.352$ · McNemar $p = 0.689$

Discordant: 56/287 (19.5%)

c = SAE yes, AAE no (SAE-directional)

Figure 2: Paired SAE–AAE contingency tables by condition. b = SAE no, AAE yes; c = SAE yes, AAE no. Discordant cells highlighted.

after the hedge, presenting fabricated content as though it had been sourced. Among hallucinated fabricated-prompt responses, 40–43% open with a hedge and then break it (119/289 across SAE+AAE); 17% have no hedge whatsoever; only a minority maintain hedge scope across the entire response. Representative excerpt:

“The response opens with a brief acknowledgment (‘I couldn’t find a publicly accessible full text...’), but it immediately

pivots to describing ‘the kind of research design and methods Patel & Romero (2022) used’ – directly attributing the methodological content to the specific fictional paper rather than maintaining clear reconstructive framing.”

Theme 2: Tacit Ratification of Ambiguity. In the underspecified condition, prompts deliberately omit identifying detail (last-name

only, partial title, missing year, partial author list). The model very rarely flags this ambiguity. Instead it silently picks one disambiguation – sometimes correct, often not – and proceeds with paper-specific confidence. Across underspecified hallucinations, 19–24% of evidence entries explicitly note that the model “never flags the ambiguity” or “tacitly ratifies” the partial citation, and an additional 17–19% involve outright author or first-name errors (F1) where the model commits to a wrong identity rather than asking. Representative excerpt:

“The response confidently attributes the paper to ‘John R. Anderson and collaborators’ and grounds it in ‘Anderson’s ACT-R framework.’ Ground truth: the paper’s lead author is Reid McIlroy-Young; Ashton Anderson is the last author and is not John R. Anderson (the ACT-R cognitive scientist).”

Tacit ratification is dialect-portable: the same disposition surfaces in both SAE and AAE underspecified responses, but the specific disambiguation chosen sometimes differs by dialect, contributing directly to the case-level disagreement quantified by Cohen’s κ in §4.4.

Theme 3: Real-Authority Anchoring. A distinct and high-severity failure mode is the laundering of fabricated content through real scholar identities. When a fictional citation gestures toward a name the model recognizes, the response does not just borrow the name – it borrows the recognizable scholar’s research program, vocabulary, and authority, and grafts these onto the fictional paper. The most striking case attaches Mizuko Ito (the real media scholar) to a fictional paper about adolescent media use; another attaches John R. Anderson and ACT-R to a chess-behavior paper that has no relation to either. F3a (real-scholar attached to fictional paper) appears in 3–4% of fabricated-prompt hallucinations (10/289), perfectly symmetric across dialects (5 SAE, 5 AAE).

Theme 4: Fabricated Specificity. 96–99% of hallucinated fabricated-prompt responses (282/289) contain at least one piece of fabricated specificity – a named methodology, a labeled finding, a numbered taxonomy, a participant count, an interaction effect. Specificity does the rhetorical work of grounding: the more concrete the detail, the more the response reads as sourced rather than reconstructed. Severity-2 (“moderately specific fabricated content”) accounts for ~95% of fabricated hallucinations; severities 3–4 (highly specific or real-scholar-anchored) account for the remaining ~5%. Representative excerpt:

“The response presents detailed, paper-specific content as confirmed fact – ‘Walters was interested in two main things: (1) Perceived scarcity... (2) Deadline acceleration.’ ‘Students’ writing behavior was tracked over time,’ ‘Most students delay writing until time feels scarce.’”

Theme 5: Confidence-Outcome Decoupling. Stated confidence does not track factual accuracy in a user-actionable way. Confidence drops only modestly when the model is hallucinating – 75–87% of hallucinated responses still receive a High-confidence label – and there is no meaningful Low-confidence band (one Low in 1,200 rows). The effect is statistically significant (χ^2 , $p < .05$ in every cell) but operationally weak: a user inspecting the model’s stated certainty cannot reliably tell hallucinated responses from grounded ones. This theme is mechanistic rather than thematic in the strict

RTA sense, but it appears repeatedly in the evidence column as an annotator observation about why the failure modes are dangerous: the model does not externalize its uncertainty in a usable form.

Cross-theme synthesis. Themes 1, 2, and 4 jointly describe a single underlying disposition: the model is built to produce specific, attribution-rich text, and its hedging mechanisms are decorative rather than scope-controlling. Theme 3 names the highest-severity expression of that disposition – borrowed authority laundering an unverified claim. Theme 5 observes that the model’s stated confidence does not differentiate hallucinated from grounded outputs strongly enough to function as a user-facing safety net. The themes are dialect-portable: they describe the model’s failure mode in both SAE and AAE, with dialect-induced variation showing up in *which* prompts trigger the disposition rather than in *whether* the disposition exists.

5 Discussion

5.1 Interpreting the (non-)effect

Across 573 paired items, AAE prompts elicited a hallucination 2.9 percentage points more often than SAE prompts – consistently directional across both source conditions, but not statistically distinguishable from chance under McNemar. The effect was somewhat larger in the fabricated condition (+4.5 pp) than the underspecified condition (+1.4 pp), which is consistent with a story in which dialect matters more when the model has full latitude to hedge (and to choose whether to hedge) than when it is constrained by the existence of a real referent. Reassuringly, the most severe error class (F3a) occurs at the same rate in both dialects (5 instances each in fabricated), suggesting that the dialect effect, if real, does not concentrate in the highest-severity failures.

We are wary of overinterpreting a directional non-significant result. With $n=573$ paired items and a baseline rate of ~64%, we are powered to detect roughly a 7–8 pp effect at $\alpha=0.05$, 80% power; a 3 pp effect would require approximately 1,400 paired prompts. The honest reading of our pilot is therefore: *the data are consistent with a small AAE-directional effect on hallucination rate, but a much larger study is needed to resolve whether this effect is real.*

5.2 High hallucination rates in underspecified queries

Independently of the dialect comparison, GPT-5 hallucinates on roughly 80% of underspecified queries about real but obscure papers regardless of dialect. When asked to summarize a paper it has not seen, the modal behavior is to confidently confirm whatever information is provided by the user, potentially fabricating methodology, findings, and arguments rather than to hedge or abstain. This has direct implications for students using LLMs in literature review: the failure mode is not rare or edge-case but is the expected outcome for any citation that falls outside the model’s training coverage.

5.3 Divergent case-by-case findings

Per §4.4, Cohen’s $\kappa = 0.28$ –0.35 between SAE and AAE variants of the same prompts is fair-to-moderate at best: even when SAE and AAE produce statistically indistinguishable aggregate hallucination rates, the model disagrees with itself by dialect on 19–36%

of individual prompts. This deserves future exploration: equivalent accuracy is not equivalent service if reliability on any given query is dialect-dependent. This is invisible to standard aggregate-accuracy reporting and would not be detected by averaged benchmark metrics. However, we believe it is more likely that this is simply due to the inherent stochasticity of LLM outputs, as we anecdotally observed variation in LLM responses to the same prompt without any dialect changes.

5.4 Discussion of RTA

The reflexive thematic analysis reframes the failure mode at the response level. Four things follow. First, hedges in this domain are not load-bearing – 40%+ of hallucinated fabricated responses begin with disclaimers that are then immediately violated by direct attribution (Theme 1). User-facing safety-by-disclaimer is therefore an unreliable mitigation surface. Second, when prompts are ambiguous, the model treats ambiguity as ratified (Theme 2): rather than asking which paper or which author is meant, it silently selects one and continues, which is the proximate mechanism of the entity errors (F1) that dominate the underspecified condition. Third, the rare but high-severity F3a real-authority anchoring (Theme 3) makes hallucinations harder for non-experts to detect because the borrowed authority’s research program is internally consistent with the laundered claim. Fourth, fabricated specificity (Theme 4) – named methodologies, numbered findings, participant counts – does the rhetorical work of grounding, making responses read as sourced rather than reconstructed. Themes 1, 2, 3, and 4 collectively describe a model trained to produce specific, attribution-rich text whose hedging is rhetorical rather than epistemic.

5.5 Connection to Hofmann et al.

Hofmann et al. (2024) [10, 11] document a strong implicit-valence effect of dialect; we document at most a weak rate effect on a different outcome (hallucination). One reading: dialect-conditioned biases may surface more strongly in tasks where the model is free to make personal judgments about the speaker than in tasks where it is asked about a third-party object (a paper). A second reading, not mutually exclusive: the kind of bias Hofmann et al. identify shows up in the *content* of hallucinations, which we have not yet annotated for valence in this paper. The natural follow-up is to extend this pipeline to a domain (police transcripts, medical complaints) where hallucinated content has clear social valence, and to annotate *what* is fabricated rather than only *whether* something is fabricated.

6 Limitations

Translation quality. Multi-VALUE is rule-based and sometimes produces register-flattened AAE. 16/600 translations required hand-patching for syntax. Natural production or validation by AAE speakers would be preferable and would also include prosodic and discourse-level features rule-based systems miss; unfortunately, this was out of our scope.

Stochasticity and single-shot inference. Each prompt was sent once at default temperature. A more rigorous design would sample $k > 1$ completions per prompt to disentangle dialect effects from sampling noise.

Single model, single domain. We audited GPT-5 only, on academic citation queries only. Generalization to other models (Claude, Gemini, Llama) and other domains (police transcripts, medical questions) is the natural next step.

Judge bias. We use Anthropic’s Claude as the judge for OpenAI’s GPT-5; we have no reason to expect cross-vendor bias on this task, but we have not formally tested for it. Specifically, the concern would be if Claude and GPT-5 make hallucinations on the same prompt/responses. Our 80-item human validation points towards “no”, but was not conducted formally.

Researcher-constructed themes. Reflexive thematic analysis foregrounds the researcher’s analytic role: themes are constructed rather than discovered. Another analyst might produce overlapping but not identical themes from the same evidence column, and our regex-based prevalence anchors capture explicit annotator phrasing rather than every paraphrased instance of a pattern. We position the RTA as descriptive rather than confirmatory.

Power. Most directly, our $n=573$ paired items leave us underpowered to detect the effect size we observe. We provide a power-calibration recommendation in §7.

7 Conclusion and Future Work

We introduced a 1,200-response paired SAE/AEE dataset and a pipeline for auditing dialect-conditioned hallucination behavior in LLMs. Our pilot on GPT-5 finds a small, consistent, but non-significant AAE-directional effect on hallucination rate; a much larger but otherwise identical study is needed to confirm or rule out this effect. Beyond the rate question, we find that paired SAE-AEE agreement is only fair-to-moderate (Cohen’s $\kappa = 0.28-0.35$), so it is still possible that equal aggregate accuracy could mask case-level dialect-induced divergence. We also note an order-of-magnitude finding that is dialect-independent: GPT-5 hallucinates on roughly 80% of queries about real-but-obscure academic references, which has direct implications for student use of LLMs in literature review. A reflexive thematic analysis of judge-evidence text identifies five recurrent failure modes – *hedge-then-fabricate*, *tacit ratification of ambiguity*, *real-authority anchoring*, *fabricated specificity*, and *confidence-outcome decoupling* – that together describe a model whose rhetorical posture of certainty outpaces the evidence it actually has.

Our study should be extended in 3 ways: **(1) Power.** A power analysis on the McNemar exact test indicates that ~1,400 paired prompts would be needed to detect a 3 pp AAE-directional effect at the observed marginal hallucination rate of ~30% in fabricated cases at $\alpha=0.05$, 80% power; we used 600 due to time constraints. **(2) Models and domains.** Replication across Claude, Gemini, and open-weight Llama variants, and across at least one socially high-stakes domain such as police transcript summarization, where hallucinated content has clearer social valence. **(3) Human validation.** The 80-item judge audit, while reasonable for our pilot, is too small and informal; a ~200-item multiple-annotator study with Cohen’s κ inter-rater agreement would substantially strengthen the pipeline.

Beyond dialect analysis, our findings suggest that future work should additionally focus on documenting and mitigating the high rates of hallucination in GPT-5.

Author Contributions

Kirin DaneK: Project conceptualization and proposal writing; literature review and integration of major related works (dialect-based QoS auditing, AAE linguistics, hallucination in LLMs); full methodology design including pilot testing of AAE translation methods, 2x2 prompt design, and prompt engineering for eliciting hallucinations; development of LLM-as-judge annotation pipeline (hallucination taxonomy selection, codebook creation, ground-truth summary writing, system prompt engineering, Anthropic API implementation); SAE-to-AAE translation of all prompts with manual quality review and error correction; data collection (25% of GPT responses); statistical analysis design (McNemar’s test); paper outline and first draft; final revisions and submission.

Sri Indla: Gathered collection of related work articles, ran majority (>50%) of data collection, performed statistical analysis, supervision in project revision through project maturation, did reflexive thematic analysis

Fatiha Haider: Supervision, Data Prompt Generation and Response Collection, Analyzed Hallucination Categorization, Writing – Review & Editing.

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A Underspecification Categories

Underspecification categories are divided as following. The listed sub-categories are either applied alone or in combinations with other sub-categories in the parent category.

Author name degradation:

- First name only
- Last name only
- Common last name with no disambiguation
- Wrong first name with correct last name
- Misspelled surname
- Name transposition
- Partial author list
- Wrong coauthor
- Noncanonical author emphasis

Title degradation:

- Abbreviated title
- Paraphrased title
- Partial title
- One wrong word

- Synonym substitution
- Missing subtitle
- Shortened title that removes a key disambiguating phrase

Venue/metadata degradation:

- Wrong year
- Wrong venue
- Wrong journal
- Wrong conference
- Wrong volume or page information
- Wrong venue type
- Conflating an arXiv preprint with a later publication
- Conflating conference and journal versions
- Conflating two real citations for related versions

Conceptual misattribution:

- Attributing a method to the wrong group
- Merging two related papers into one

- Asking about a real method with a slightly wrong name
- Wrong acronym
- Reversing a paper's core claim
- Changing a crucial conceptual term
- Calling a benchmark a method, or a method a benchmark
- Merging two distinct concepts into one

Mixed degradation:

- Author name degradation combined with title degradation
- Author name degradation combined with venue/metadata degradation
- Title degradation combined with venue/metadata degradation
- Author name, title, and venue/metadata degradation combined
- Author name, title, venue/metadata, and conceptual misattribution combined